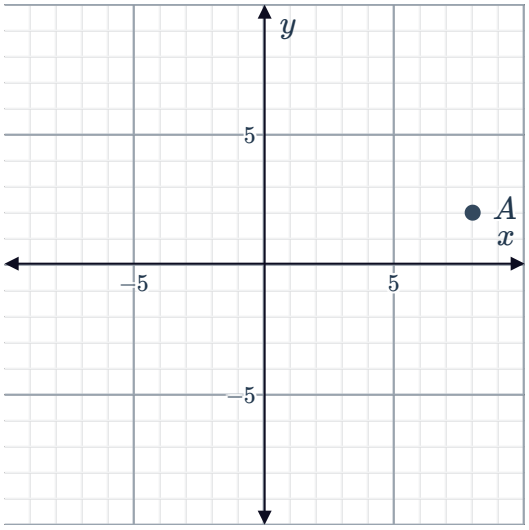
Compositions of transformations

1 Point A on the coordinate plane is to undergo two successive transformations.

a Plot A', which is the result when A is translated 3 units left.

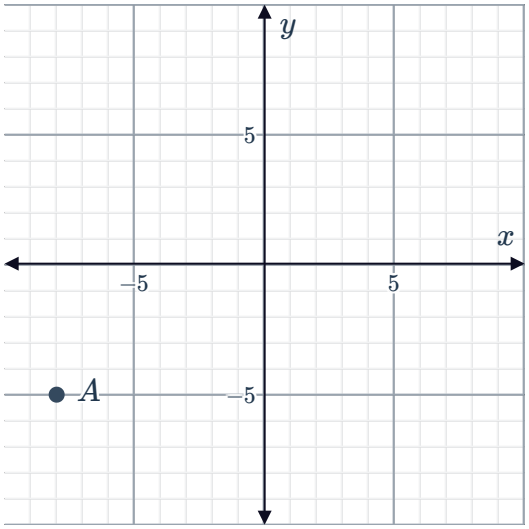
b Plot A'', which is the result when A' is rotated 90° clockwise about the origin.



2 Point A on the coordinate plane is to undergo two successive transformations.

a Plot A', which is the result when A is translated 2 units upwards.

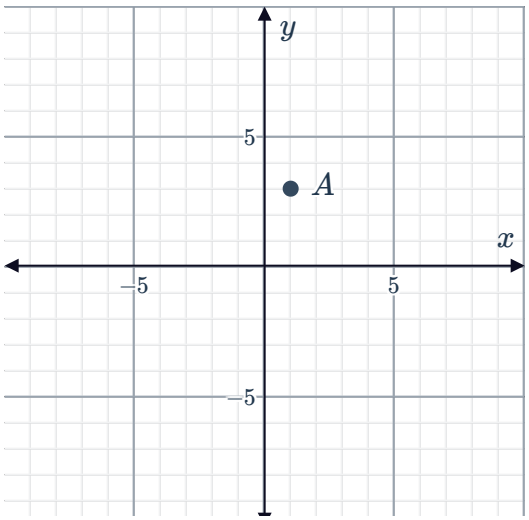
b Plot A'', which is the result when A' is rotated 90° clockwise about the origin.



3 Point A on the coordinate plane is to undergo two successive transformations.

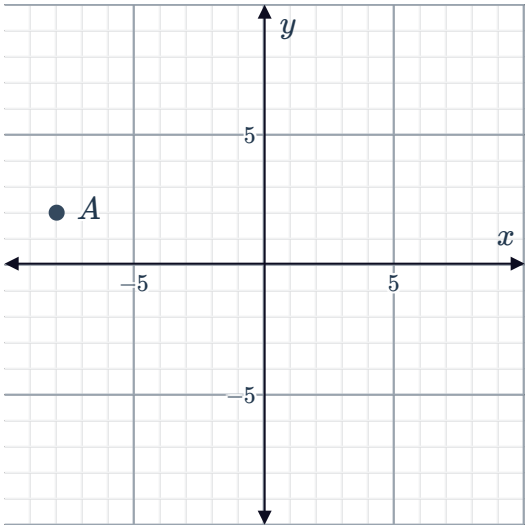
a Plot A', which is the result when A is translated 3 units right and 5 units down.

b Plot A'', which is the result when A' is rotated 90° clockwise about the origin.

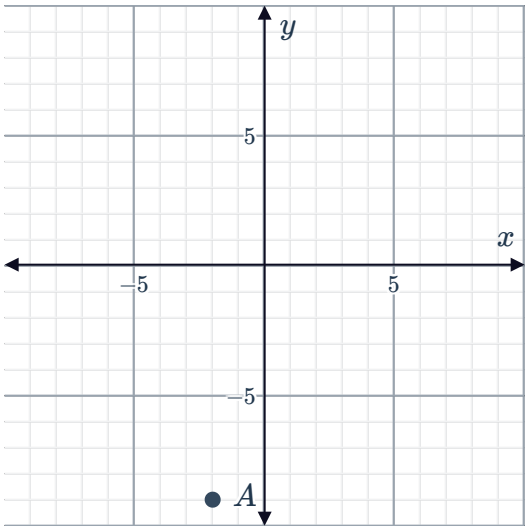


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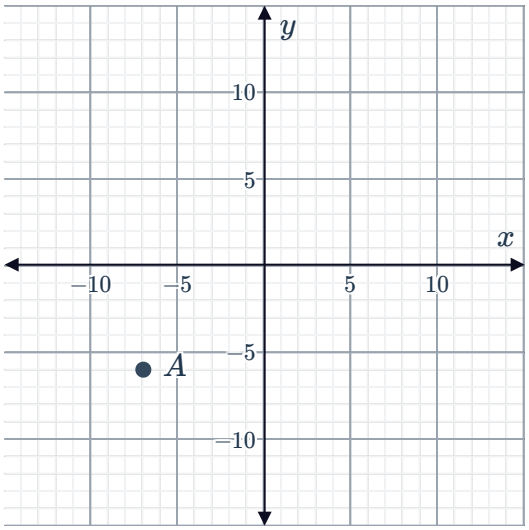
4 For the point on the coordinate plane, plot A' that will result from a translation of 5 units right followed by a rotation of 90° counterclockwise about the origin.



5 For the point on the coordinate plane, plot A' that will result from a translation of 3 units upwards followed by a rotation of 90° counterclockwise about the origin.

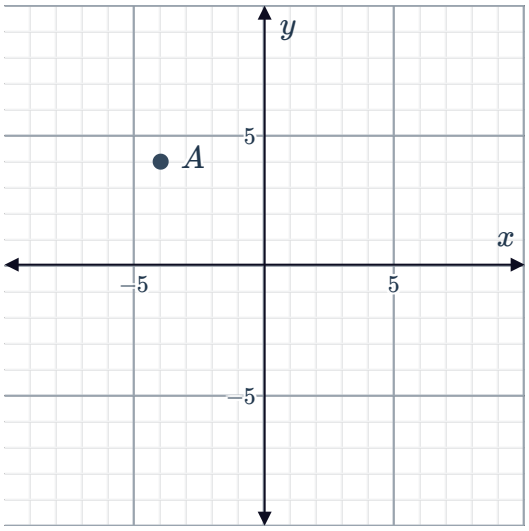


6 For the point on the coordinate plane, plot A' that will result from a translation of 5 units left and 5 units down, followed by a rotation of 90° counterclockwise about the origin.



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7 A is marked on the following coordinate plane.



a If A is translated 8 units down to make B, determine the coordinates of B.

b Across what line could A be reflected so that it overlaps B?

8

a For any point (a, b) on the coordinate plane, determine the coordinates of the point after a rotation of 180° counterclockwise about the origin.

b True or False: A rotation of 180° counterclockwise about the origin is the same as reflecting the point across both the x- and y-axes.

9 $A\,(-9, 7)$ is rotated 90° clockwise about the origin to form B.

a Determine the coordinates of B.

b B is then translated to the left so that it lies on the same vertical line as point A. How many units to the left is point B translated?

10 Determine the coordinates of the transformed point for each of the following:

a The point $(-9, -1)$ undergoes translation of 2 units down followed by a rotation of 90° clockwise about the origin.

b The point $(9, 2)$ undergoes a translation of 2 units up followed by a rotation of 90° counterclockwise about the origin.

c The point on the coordinate plane $(9, -2)$ undergoes a translation of 5 units left and 5 units up, followed by a rotation of 90° clockwise about the origin.

d The point $(-2, 6)$ undergoes a translation of 5 units right and 4 units up, followed by a rotation of 90° counterclockwise about the origin.

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11 $A\,(6, -1)$ is first rotated 180° about the origin, and then it is reflected across the x-axis. This produces the point A'.

a Determine the coordinates of A'.

b Determine whether the following transformations also takes A to A'.

i A reflection across the y-axis.

ii A 90° clockwise rotation about the origin.

iii A vertical translation.

iv There is no other way to describe the transformation.

12 Consider the two points, A and B on the coordinate plane.

a Plot the points that would result when A and B are rotated 180° about the origin.

b Plot the points that would result when A and B are first rotated 180° about the origin, and then reflected across the x-axis.

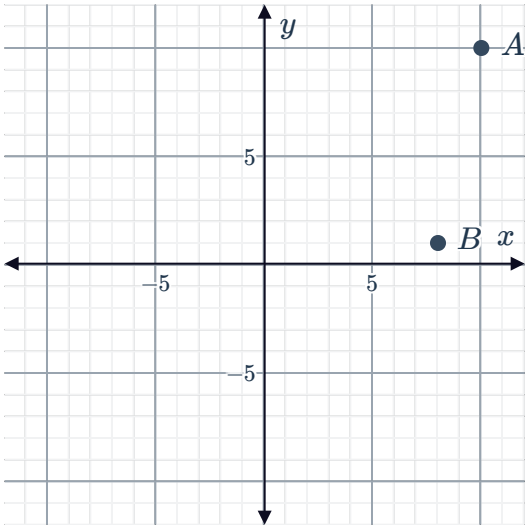
c Determine whether the following could be use to describe the transformation in part (b).

i There is no other way to describe the transformation.

ii A reflection across the y-axis.

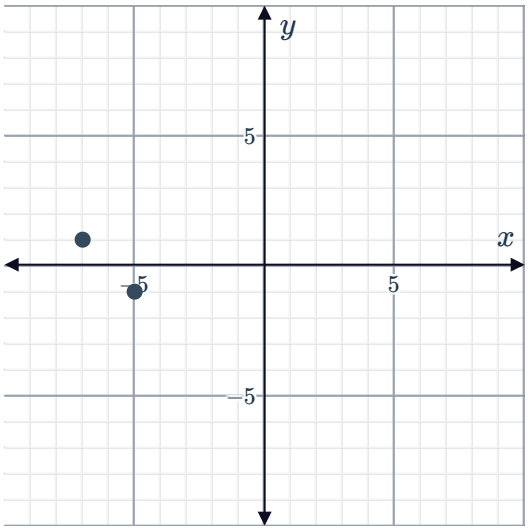
iii A horizontal translation.

iv A vertical translation.



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13 Consider the two points on the coordinate plane.



a Plot the points that would result when the original points are rotated 180° about the origin and reflected across the y-axis.

b Determine whether the following could describe the transformation.

i A horizontal translation.

ii There is no other way to describe the transformation.

iii A vertical translation.

iv A reflection across the x-axis.

14 Consider the points $A\,(-10, -8)$ and $B\,(-7, -7)$.

a Determine the coordinates of A' and B' that would result when the original points are rotated 180° about the origin and reflected across the y-axis.

b Determine whether the following could also describe the transformation.

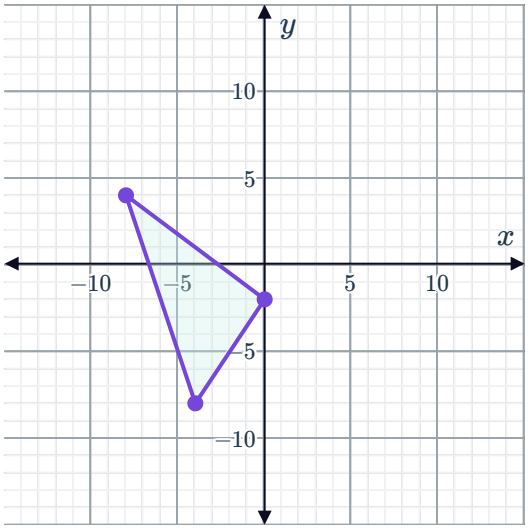
i There is no other way to describe the transformation.

ii A reflection across the x-axis.

iii A vertical translation.

iv A horizontal translation.

15 The given triangle is to undergo two transformations.

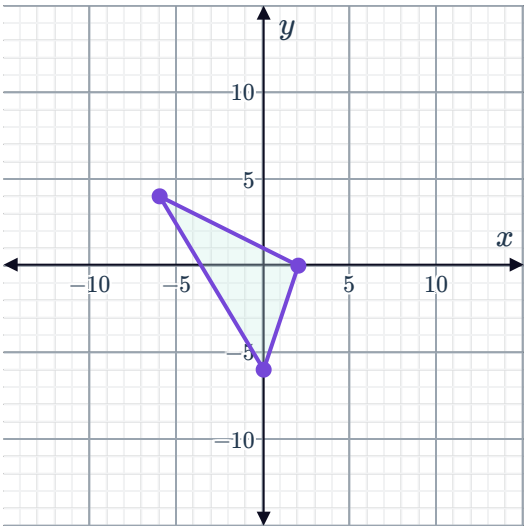


a Plot the triangle that results from reflecting the given triangle across the x-axis.

b Plot the triangle that results when the triangle from part b is translated 4 units right.

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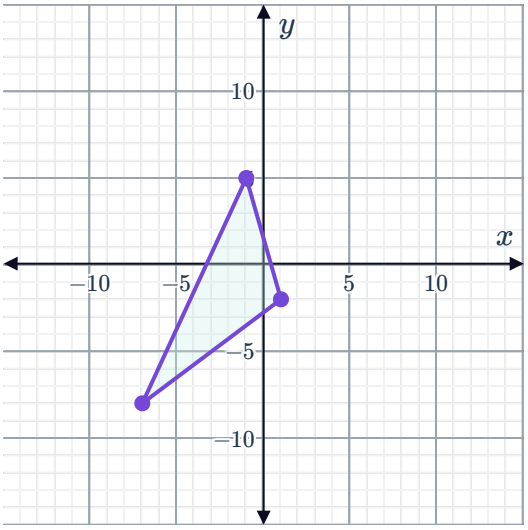
16 Plot the triangle that will result when the given triangle is reflected across the x-axis and translated 3 units to the right.



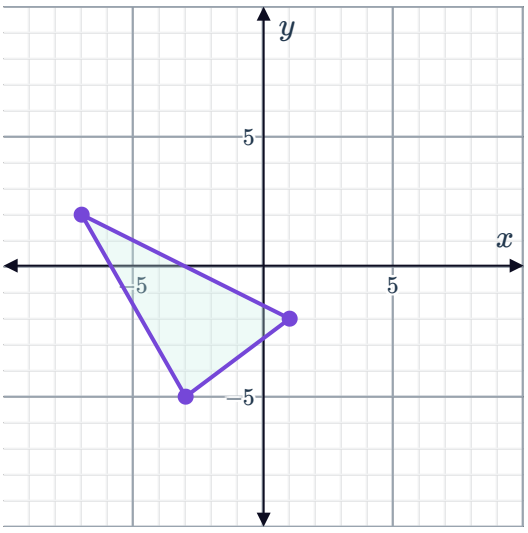
17 The given triangle is to undergo two transformations.

a Plot the triangle that results from reflecting the given triangle across the y-axis.

b Plot the triangle that results when the triangle from part b is translated 3 units right.



18 Plot the triangle that will result when the original triangle is reflected across the y-axis and translated 3 units down.

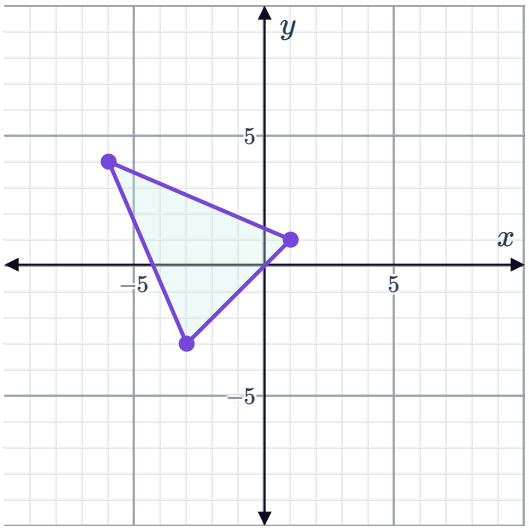


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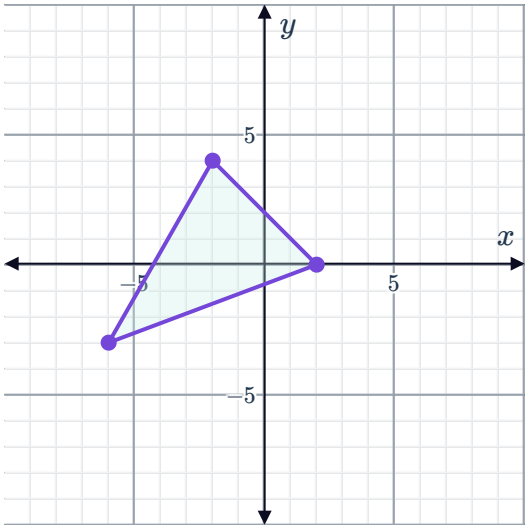
19 The given triangle is to undergo two transformations.

a Plot the triangle that results from reflecting the given triangle across the x-axis.

b Plot the triangle that results when the triangle from part b is translated 2 units right and and 4 units up.



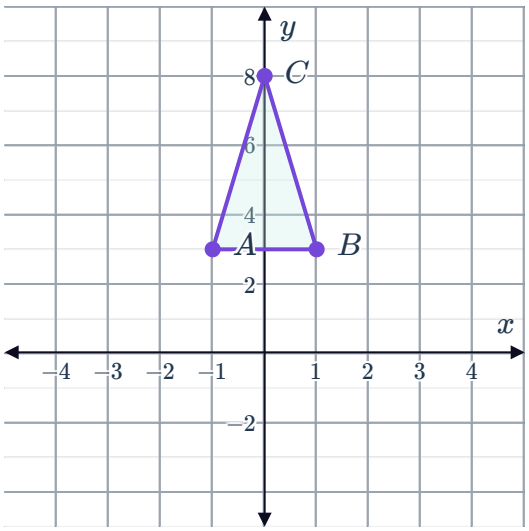
20 Plot the triangle that will result when the original triangle is reflected across the y-axis, and translated 2 units left and 3 units down.



21 $\triangle ABC$ is to undergo two separate reflections.

a Plot $\triangle A'B'C'$, the result of reflecting $\triangle ABC$ across the line $y = x$.

b Now suppose $\triangle A'B'C'$ is reflected across the y-axis to form $\triangle A''B''C''$. What single transformation would overlap $\triangle ABC$ onto triangle $\triangle A''B''C''$?



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22 Points $A\,(-7,5)$, $B\,(-5,-8)$, and $C\,(10,-10)$ are the vertices of a triangle.
Determine the coordinates of A', B', and C' that result from reflecting the triangle across the x-axis and translating it 4 units right.

23 Points $A\,(-10,4)$, $B\,(5,-5)$, and $C\,(5,-2)$ are the vertices of a triangle.
Determine the coordinates A', B', and C' that result from reflecting the triangle across the x-axis and translating it 3 units down.

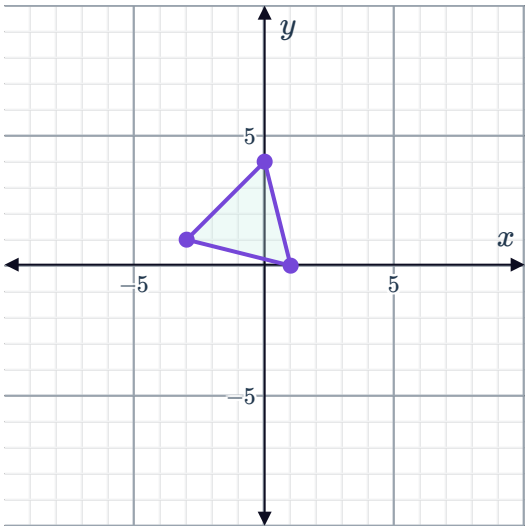
24 Points $A\,(4,-9)$, $B\,(4,3)$, and $C\,(-8,2)$ are the vertices of a triangle.
Determine the coordinates A', B', and C' that result from reflecting the triangle across the y-axis and translating it 5 units right.

25 Points $A\,(8,-4)$, $B\,(5,6)$, and $C\,(9,9)$ are the vertices of a triangle.
Determine the coordinates A', B', and C' that result from reflecting the triangle across the y-axis and translating it 3 units up.

26 Points $A\,(-1,-9)$, $B\,(-6,-8)$, and $C\,(8,9)$ are the vertices of a triangle.
Determine the coordinates A', B', and C' that result from reflecting the triangle across the x-axis, and translating it 5 units right and 3 units down.

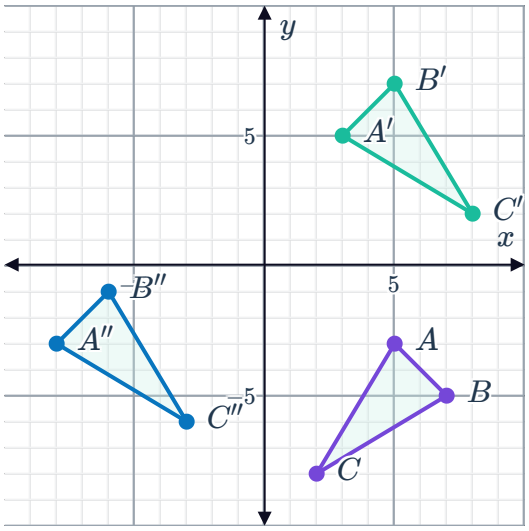
27 Points $A\,(-5,-7)$, $B\,(4,4)$, and $C\,(9,1)$ are the vertices of a triangle.
Determine the coordinates A', B', and C' that result from reflecting the triangle across the y-axis, and translating it 3 units right and 5 units up.

28 Consider the triangle shown on the coordinate grid.
Plot the triangle that results when the first triangle undergoes the transformation $(x,y) \to (x-3, y+4)$.



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29 Determine whether the following sequences of transformations map $\triangle ABC$ to $\triangle A''B''C''$.



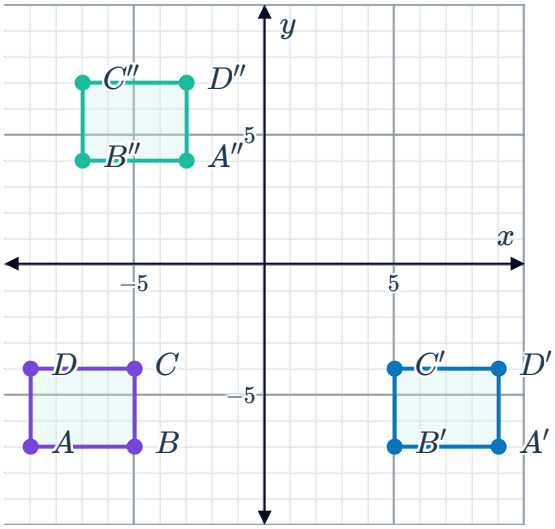
a A reflection across the x-axis, and a translation 11 units left and 8 units down.

b A translation 11 units left and 8 units down, and a reflection across the x-axis.

c A translation 11 units left and 8 units down, and a rotation by 90° counterclockwise about the origin.

d A rotation by 90° counterclockwise about the origin, and a translation 11 units left and 8 units down.

30 Determine whether the following sequences of transformations map quadrilateral $ABCD$ to quadrilateral $A''B''C''D''$.



a A translation 12 units left and 11 units up, and a rotation by 270° clockwise about the origin.

b A rotation by 270° clockwise about the origin, and a translation 12 units left and 11 units up.

c A translation 12 units left and 11 units up, and a reflection across the y-axis.

d A reflection across the y-axis, and a translation 12 units left and 11 units up.

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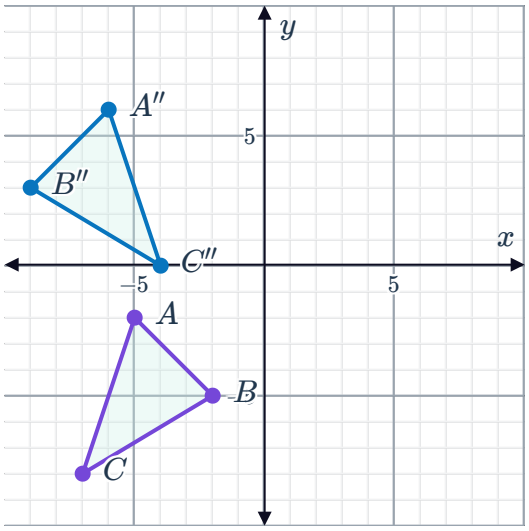
31 Determine whether the following sequences of transformations map $\triangle ABC$ to $\triangle A''B''C''$.

a A rotation by 180° counterclockwise about the origin, and a reflection across the x-axis.

b A reflection across the x-axis, and a translation 11 units right and 8 units up.

c A translation 11 units right and 8 units up, and a reflection across the y-axis.

d A reflection across the y-axis, and a rotation by 180° counterclockwise about the origin.



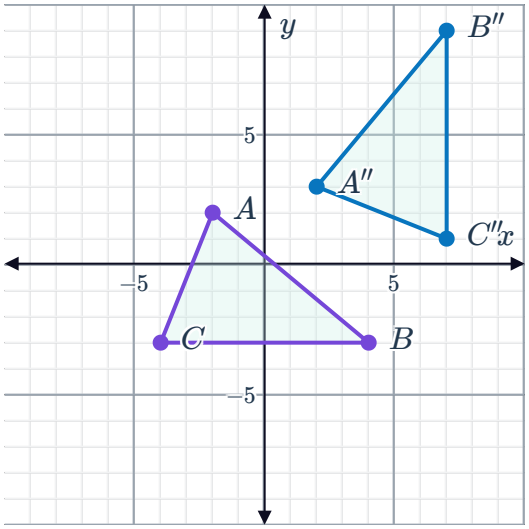
32 Consider the following diagram.
Complete the tables below to identify the change in coordinates going through the sequence of transformations from $\triangle ABC$ to $\triangle A''B''C''$.

a A translation 5 units to the right and 4 units down.

$A(-2,2)$	$\rightarrow$	$A'(\square,\square)$
$B(4,-3)$	$\rightarrow$	$B'(\square,\square)$
$C(-4,-3)$	$\rightarrow$	$C'(\square,\square)$

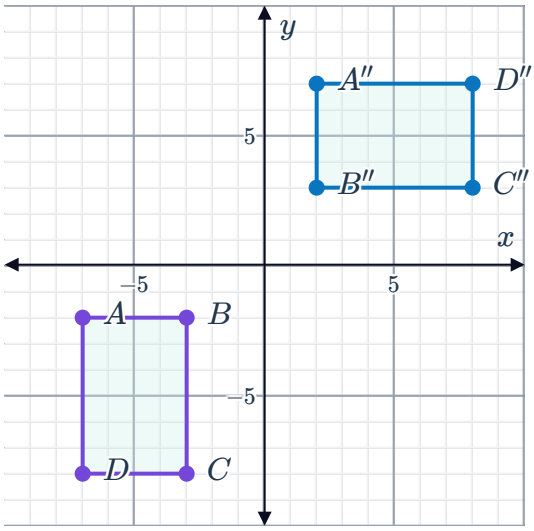
b A rotation 270° clockwise about the origin.

$A'(\square,\square)$	$\rightarrow$	$A''(\square,\square)$
$B'(\square,\square)$	$\rightarrow$	$B''(\square,\square)$
$C'(\square,\square)$	$\rightarrow$	$C''(\square,\square)$



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33 Consider the following diagram.
Complete the tables below to identify the change in coordinates going through the sequence of transformations from quadrilateral $ABCD$ to quadrilateral $A''B''C''D''$.



a A rotation by 90° clockwise about the origin.

$A(-7,-2)$	$\rightarrow$	$A'(\square,\square)$
$B(-3,-2)$	$\rightarrow$	$B'(\square,\square)$
$C(-3,-8)$	$\rightarrow$	$C'(\square,\square)$
$D(-7,-8)$	$\rightarrow$	$D'(\square,\square)$

b A reflection across the y-axis.

$A'(\square,\square)$	$\rightarrow$	$A''(\square,\square)$
$B'(\square,\square)$	$\rightarrow$	$B''(\square,\square)$
$C'(\square,\square)$	$\rightarrow$	$C''(\square,\square)$
$D'(\square,\square)$	$\rightarrow$	$D''(\square,\square)$

34 Consider the quadrilateral with vertices at $(5,4)$, $(0,4)$, $(0,-1)$, and $(5,-1)$, and the quadrilateral with vertices at $(4,5)$, $(4,0)$, $(-1,0)$, and $(-1,5)$.
Determine whether the following sequences of transformations maps the first quadrilateral to the second quadrilateral.

a Rotated 270° clockwise about the origin and reflected across the x-axis.

b Reflected across the x-axis and rotated 270° clockwise about the origin.

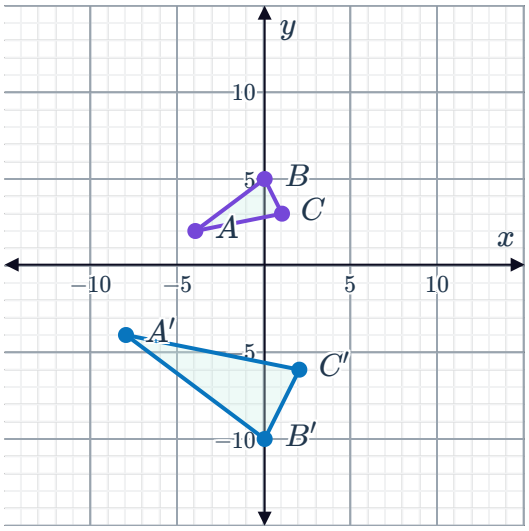
c Reflected across the x-axis and translated 2 units right and 3 units up.

d Translated 2 units right and 3 units up and reflected across the x-axis.

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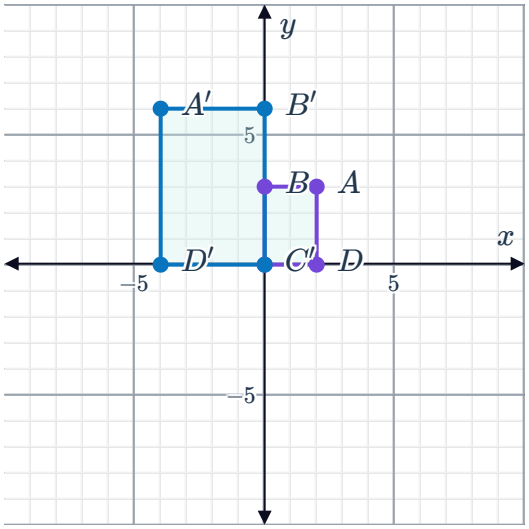
35 Determine whether the following sequences of transformations maps $\triangle ABC$ to $\triangle A'B'C'$.

- **a** Translated 3 units right and4 units down, and dilated by a factor of 2 with the origin as the center of dilation.
- **b** Rotated 90° clockwise around the origin and reflected about the x-axis.
- **c** Reflected about the x-axis and dilated by a factor of 2 with the origin as the center of dilation.
- **d** Rotated 90° clockwise around the origin and dilated by a factor of 2 with the origin as the center of dilation.



36 Determine whether the following sequences of transformations maps quadrilateral $ABCD$ to quadrilateral $A'B'C'D'$.

- **a** Reflected about the y-axis and rotated 180° clockwise around the origin.
- **b** Translated 3 units left and 2 units down, and dilated by a factor of 2 with the origin as the center of dilation.
- **c** Rotated 180° clockwise around the origin and dilated by a factor of 2 with the origin as the center of dilation.
- **d** Dilated by a factor of 2 with the origin as the center of dilation and reflected about the y-axis.



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37 Consider the triangle with vertices at $A\,(-4,2)$, $B\,(-1,5)$, and $C\,(1,3)$, and the triangle with vertices at $A'\,(8,4)$, $B'\,(2,10)$, and $C'\,(-2,6)$.
Determine whether the following sequences of transformations maps the first triangle to the second triangle.

- **a** Dilated by a factor of 2 with the origin as the center of dilation and rotated 180° counterclockwise around the origin.
- **b** Rotated 180° counterclockwise around the origin and translated 5 units right and 4 units down.
- **c** Dilated by a factor of 2 with the origin as the center of dilation, and translated 5 units right and 4 units down.
- **d** Reflected about the y-axis and dilated by a factor of 2 with the origin as the center of dilation.

38 Consider the quadrilateral with vertices at $(3,2)$, $(-1,2)$, $(-1,-3)$, and $(3,-3)$, and the quadrilateral with vertices at $(-6,9)$, $(-6,-3)$, $(9,-3)$, and $(9,9)$.
Determine whether the following sequences of transformations maps the first quadrilateral to the second quadrilateral.

- **a** Dilated by a factor of 3 with the origin as the center of dilation and rotated 90° counterclockwise around the origin.
- **b** Reflected about the y-axis and translated 2 units left and 3 units down.
- **c** Reflected about the y-axis and dilated by a factor of 3 with the origin as the center of dilation.
- **d** Dilated by a factor of 3 with the origin as the center of dilation, and translated 2 units left and 2 units down.

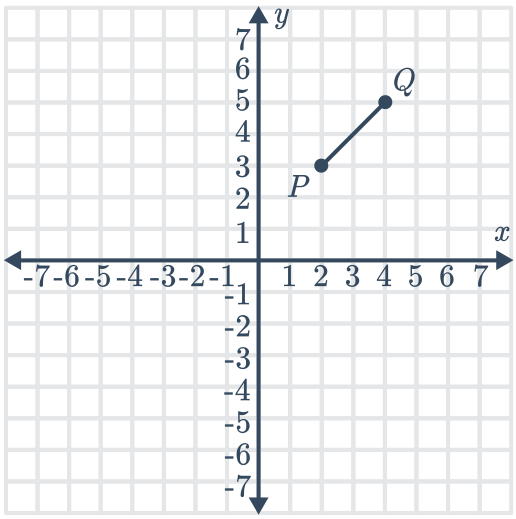
39 The quadrilateral with vertices at $(9,9)$, $(0,9)$, $(0,0)$, and $(9,0)$ is rotated 90° clockwise around the origin and dilated by a factor of 2 with the origin as the center of dilation.
Determine the new coordinates of the vertices of the quadrilateral.

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Additional questions

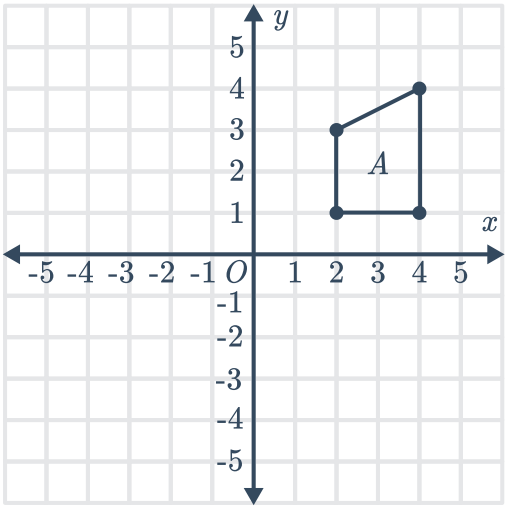
40 Perform the following sequence of transformations on $\overline{PQ}$:

- Rotate 90° counterclockwise about the origin
- Translate 5 units down



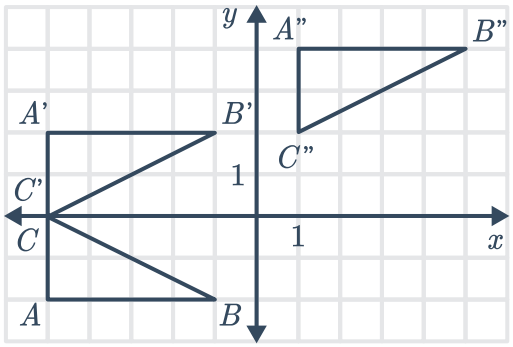
41 Perform the following sequence of transformations on Figure A:

- Reflection across the y-axis
- Rotation of 180° clockwise about the origin

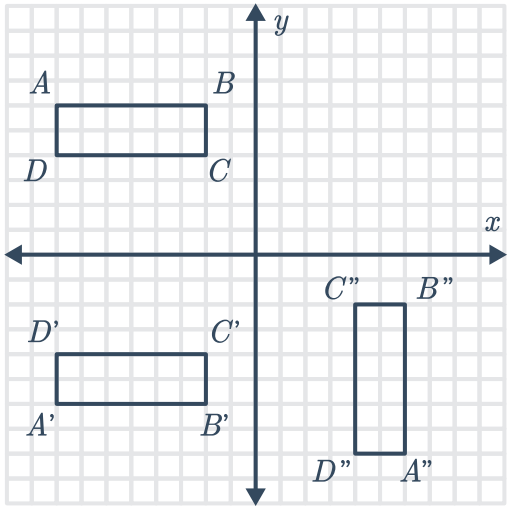


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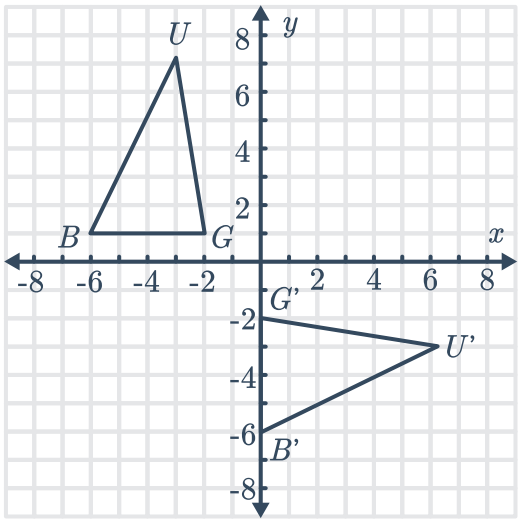
42 Describe a sequence of transformations that maps $\triangle ABC$ to $\triangle A''B''C''$:



43 Describe a sequence of transformations that maps $ABCD$ to $A''B''C''D''$:



44 Describe a sequence of transformations that maps $\triangle BUG$ to $\triangle B''U''G''$:



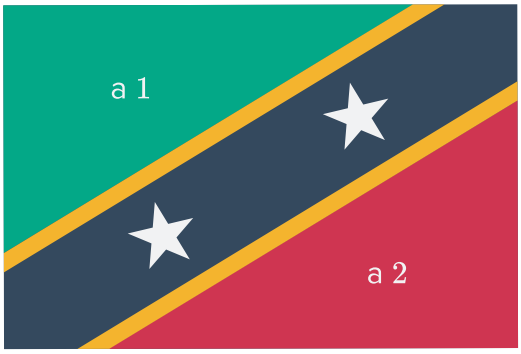
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45 The endpoints of $\overline{JK}$ are$(9, 1)$, $(10, 3)$.
Graph the image of $\overline{JK}$ after rotation by 180° about the origin then reflection across the x-axis.

46 The vertices of quadrilateral $JKLM$ are$(-4, -2)$, $(0, 8)$, $(5, 0)$, and$(5, -1)$.
Graph the image of $JKLM$ after the following sequence of transformations:

- Rotate 180° counterclockwise
- Reflect across $y = 1$

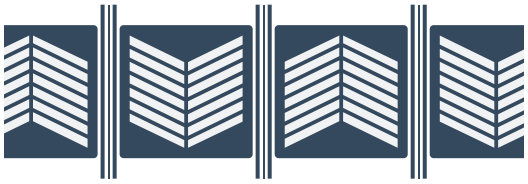
47 Here is the flag for Trinidad and Tobago:



a Describe how to map section 1 of the flag onto section 2 using a single transformation.

b Describe how to map section 1 of the flag onto section 2 using two consecutive transformations.

48 Here is pattern from a piece of Kente fabric, woven in Ghana:



Describe a sequence of transformations that could be used to recreate the pattern.

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